

Resampling Methods

Cross-Validation and the Bootstrap

Dr. Rajesh Kumar Mundotiya

Assistant Professor, Department of Computer Science
and Engineering

Lab In-Charge: MATRA Lab, Indian Institute of
Technology Bhilai



MATRA LAB

Session Roadmap

Recap & Objectives

Mathematical Prerequisites

Cross-Validation: The Problem

The Validation Set Approach

Leave-One-Out Cross-Validation

K-Fold Cross-Validation

Bias–Variance Trade-Off for K

Cross-Validation for Classification

The Bootstrap

Applied View: Cross-Validation in Code

Summary



SECTION 1

Recap & Objectives

Where We Were: Classification

From the previous session

We modelled a **categorical** response: logistic regression and LDA both estimate $\Pr(G = k \mid \mathbf{x})$ and are fit by **maximum likelihood**, giving us a decision boundary instead of a fitted line.

The gap this session fills

Every model so far — linear regression, logistic regression, LDA — has only been *evaluated on the data it was trained on*. How do we honestly estimate how well it will do on *new* data, without collecting a fresh sample? **Resampling methods** answer this.

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By the end of this session you will be able to:

1. Distinguish training error, test error, and expected prediction error.
2. Apply the validation-set, leave-one-out (LOOCV) and K -fold cross-validation procedures.
3. Explain the bias–variance trade-off that governs the choice of K .
4. Use cross-validation for both model *assessment* and model *selection*, in regression and classification.
5. Describe the bootstrap as sampling with replacement from the empirical distribution $\hat{F}$, and use it to estimate standard errors.

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Why Resampling Is the Right Next Step

Main idea

Resampling methods repeatedly draw modified versions of the training sample $\mathcal{T} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$ and refit the learning method, in order to learn something a single fit cannot tell us on its own.

What we get

- ▶ An honest estimate of test error, without a new data set.
- ▶ A way to choose model complexity or tuning parameters.
- ▶ Standard errors for statistics with no easy formula.

Two workhorses

- ▶ **Cross-validation**: repeated train/validation splits.
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Training error $\overline{\text{err}} = \frac{1}{N} \sum_i L\{y_i, \hat{f}(\mathbf{x}_i)\}$ is computed on data used to fit $\hat{f}$; it is almost always *optimistic*. Test error $\mathbb{E}_0[L\{Y_0, \hat{f}(\mathbf{x}_0)\}]$ uses a fresh $(\mathbf{x}_0, Y_0)$ and is what we actually care about.

2. The validation-set idea

If we cannot get new data, we can *pretend* some of our own data is new: hold part of $\mathcal{T}$ out, fit on the rest, and evaluate on the held-out part. This single trick underlies everything in this session.

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Splitting data into folds samples *without* replacement (each point used once). The bootstrap samples *with* replacement (a point can appear 0, 1, or many times) — a different resampling philosophy for a different question.

4. Empirical distribution $\hat{F}$

Given $\mathcal{Z} = \{z_1, \dots, z_N\}$, the empirical distribution puts mass $\hat{F}(z_i) = 1/N$ on each observed point. Bootstrap resampling is literally sampling from $\hat{F}$ in place of the unknown population distribution F .

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Model Assessment vs. Model Selection

Model assessment

How well does the fitted rule $\hat{f}$ predict new observations?

- ▶ Focus: estimate generalisation error.
- ▶ Example: estimate $\mathbb{E}\{(Y_0 - \hat{f}(\mathbf{x}_0))^2\}$.

Model selection

Which candidate model $M_1, \dots, M_L$ should be used?

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- ▶ Example: choose polynomial degree or λ .

Key link

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- ▶ Computed on data already used for fitting.
- ▶ Decreases (almost) monotonically with flexibility.
- ▶ Rewards memorisation, not generalisation.

Test error

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Teaching point

Very low training error can coexist with high test error. Cross-validation estimates test error using *only* observed data.

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The Validation Set Approach



FIGURE 5.1. A schematic display of the validation set approach. A set of n observations are randomly split into a training set (shown in blue, containing observations 7, 22, and 13, among others) and a validation set (shown in beige, and containing observation 91, among others). The statistical learning method is fit on the training set, and its performance is evaluated on the validation set.

The data are split into a training sample and a validation set.

Procedure

1. Randomly split $\mathcal{T} = \mathcal{T}_1 \cup \mathcal{V}$.
2. Fit the model on $\mathcal{T}_1$ only.
3. Predict responses in $\mathcal{V}$.
4. Compute validation error (often MSE):

$$\text{MSE}_{\mathcal{V}} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} (y_i - \hat{f}_{\mathcal{T}_1}(\mathbf{x}_i))^2.$$

Auto Example: Choosing Polynomial Degree

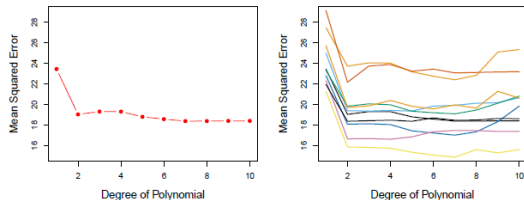


FIGURE 5.2. The validation set approach was used on the **Auto** data set in order to estimate the test error that results from predicting **mpg** using polynomial functions of **horsepower**. Left: Validation error estimates for a single split into training and validation data sets. Right: The validation method was repeated ten times, each time using a different random split of the observations into a training set and a validation set. This illustrates the variability in the estimated test MSE that results from this approach.

Question

Predict *mpg* from *horsepower*: linear, quadratic, cubic, or higher-degree polynomial?

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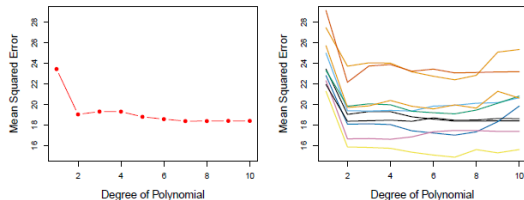


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Drawbacks of the Validation Set Approach

1. High variability

- ▶ Different random splits → different validation errors.
- ▶ The chosen model can change across splits.

2. Wasted training data

- ▶ Model trained on only part of the data.
- ▶ Validation error tends to *overestimate* the test error of a model trained on the full data.

Motivation for cross-validation

Cross-validation uses the data more efficiently and removes the dependence on a single random split.

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Leave-One-Out Cross-Validation (LOOCV)

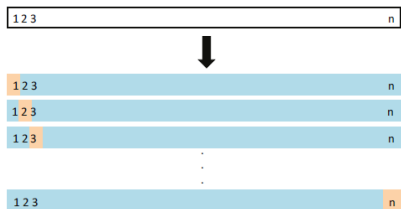


FIGURE 5.3. A schematic display of LOOCV. A set of n data points is repeatedly split into a training set (shown in blue) containing all but one observation, and a validation set that contains only that observation (shown in beige). The test error is then estimated by averaging the n resulting MSEs. The first training set contains all but observation 1, the second training set contains all but observation 2, and so forth.

Each observation is held out exactly once.

Idea

For each $i = 1, \dots, N$:

1. Remove $z_i = (\mathbf{x}_i, y_i)$.
2. Fit on $\mathcal{T}^{(-i)} = \mathcal{T} \setminus \{z_i\}$.
3. Predict y_i ; record the squared error.

Held-out error for observation i

$$\text{MSE}_i = (y_i - \hat{f}^{(-i)}(\mathbf{x}_i))^2.$$

LOOCV estimate

$$\text{CV}_{(N)} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}^{(-i)}(\mathbf{x}_i))^2. \quad (5.1)$$

- ▶ Every point is used once for validation, every model trained on $N - 1$ observations.
- ▶ No randomness: repeating LOOCV always gives the same answer.

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Advantages and Disadvantages of LOOCV

Advantages

- ▶ Training sets of size $N - 1$: close to the full-data fit.
- ▶ No randomness in the split.
- ▶ Uses every observation for validation exactly once.

Disadvantages

- ▶ Naively needs N model fits — expensive for slow methods.
- ▶ Often has *higher variance* than 5- or 10-fold CV.

The LOOCV Shortcut for Least Squares

One fit is enough

For ordinary least squares, LOOCV can be computed from a single full-data fit:

$$\text{CV}_{(N)} = \frac{1}{N} \sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2. \quad (5.2)$$

- ▶ $\hat{y}_i$: fitted value from the model trained on *all* data.
- ▶ $h_i = \mathbf{H}_{ii}$: leverage of observation i , from $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$.
- ▶ High-leverage points get a larger residual correction.

Why it works

Since $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a linear smoother, the leave-one-out residual equals $\frac{y_i - \hat{y}_i}{1 - h_i}$: the ordinary residual is too small because z_i helped fit $\hat{y}_i$; dividing by $1 - h_i$ removes that self-influence.

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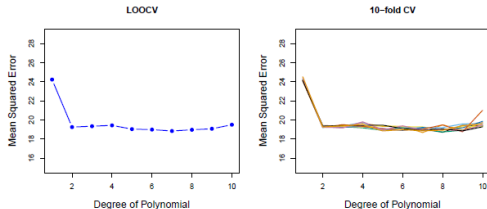


FIGURE 5.4. Cross-validation was used on the **Auto** data set in order to estimate the test error that results from predicting **mpg** using polynomial functions of **horsepower**. Left: The LOOCV error curve. Right: 10-fold CV was run nine separate times, each with a different random split of the data into ten parts. The figure shows the nine slightly different CV error curves.

Reading the figure

- ▶ Left: LOOCV error curve.
- ▶ Right: repeated 10-fold CV curves.
- ▶ Error drops sharply from degree 1 to 2.
- ▶ Higher degrees: no stable further gain.

Conclusion

A quadratic model for *mpg* vs. *horsepower* is enough.

K -Fold Cross-Validation

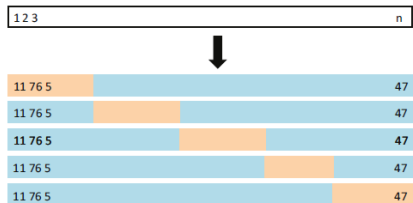


FIGURE 5.5. A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.

5-fold cross-validation.

Main idea

Randomly divide the data into K non-overlapping folds.

1. Hold out one fold.
2. Train on the remaining $K - 1$ folds.
3. Compute error on the held-out fold.
4. Repeat until every fold has been held out once.

K -Fold CV Formula

Fold-level errors

Partition $\mathcal{T} = C_1 \cup \dots \cup C_K$; let $\kappa(i)$ be the fold containing observation i .

Cross-validation estimate

$$\text{CV}_{(K)} = \frac{1}{N} \sum_{i=1}^N L\{y_i, \hat{f}^{(-\kappa(i))}(\mathbf{x}_i)\} \stackrel{\text{sq. loss}}{=} \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}^{(-\kappa(i))}(\mathbf{x}_i))^2. \quad (5.3)$$

Important special case

LOOCV is K -fold CV with $K = N$.

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Typical Choices: $K = 5$ or $K = 10$

Why not always LOOCV?

- ▶ Requires N fits — slow for complex models.
- ▶ Can have high variance.

Why 5- or 10-fold?

- ▶ Only 5 or 10 fits.
- ▶ Good practical bias–variance compromise.

Practical rule

Use 5- or 10-fold CV as a default unless data size, model cost, or the evaluation design suggests otherwise.

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SECTION 7

Bias–Variance Trade-Off for K

Bias–Variance Trade-Off for K -Fold CV

Bias side

- ▶ Validation set trains on $\sim N/2$: highest bias.
- ▶ LOOCV trains on $N - 1$: lowest bias.
- ▶ 5- or 10-fold CV sits between these extremes.

Variance side

- ▶ LOOCV fits models on nearly identical training sets $\rightarrow$ highly *correlated* errors $\rightarrow$ higher variance when averaged.
- ▶ 5- or 10-fold CV uses less-overlapping training sets, often lowering variance.

Method	Training size per fit	# fits	Main issue
Validation set	$\sim N/2$	1	high variability, high bias
LOOCV	$N - 1$	N	high computation, high variance
5/10-fold CV	$(K-1)N/K$	K	good practical compromise

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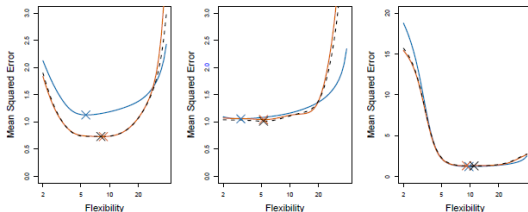


FIGURE 5.6. True and estimated test MSE for the simulated data sets in Figures 2.9 (left), 2.10 (center), and 2.11 (right). The true test MSE is shown in blue, the LOOCV estimate is shown as a black dashed line, and the 10-fold CV estimate is shown in orange. The crosses indicate the minimum of each of the MSE curves.

Reading the figure

- ▶ Blue: true test MSE.
- ▶ Black dashed: LOOCV estimate.
- ▶ Orange: 10-fold CV estimate.
- ▶ CV may over- or under-estimate the exact value ...
- ▶ ... but it usually identifies the right complexity.

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For *assessment* we care about the estimated value $\widehat{\text{Test Error}}$; for *selection* we care about the location $\arg \min_m \text{CV}(m)$. CV need not be perfect at the former to be very useful for the latter.

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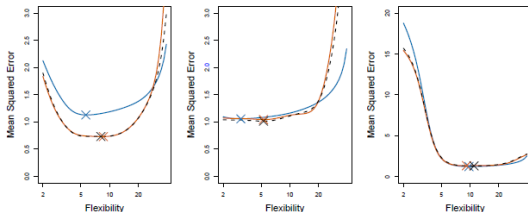


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Same idea, different loss

Setting	Response	CV error measure
Regression	quantitative	MSE
Classification	qualitative	misclassification rate

LOOCV classification error

$$CV_{(N)} = \frac{1}{N} \sum_{i=1}^N \text{Err}_i, \quad \text{Err}_i = \mathbb{1}(y_i \neq \hat{y}_i). \quad (5.4)$$

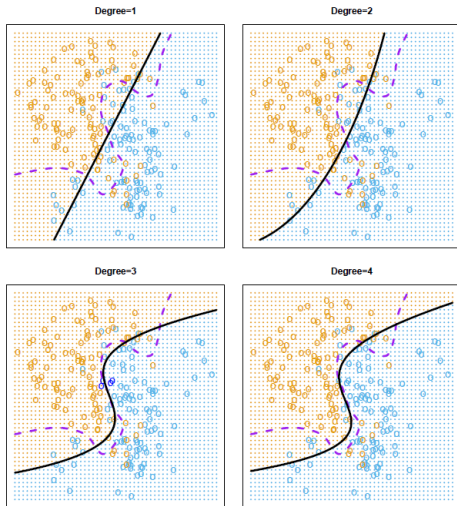
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Classification Example: Logistic Regression Boundaries



Reading the figure

- ▶ Purple dashed: Bayes decision boundary.
- ▶ Black: fitted logistic regression boundary.
- ▶ Higher-degree polynomial terms:

$$\log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4$$

give more flexible, curved boundaries.

- ▶ Degree 3 helps a lot; degree 4 is not clearly better.

FIGURE 5.7. Logistic regression fits on the two-dimensional classification data displayed in Figure 2.13. The Bayes decision boundary is represented using a purple dashed line. Estimated decision boundaries from linear, quadratic, cubic

Classification CV: Training Error vs. Test Error

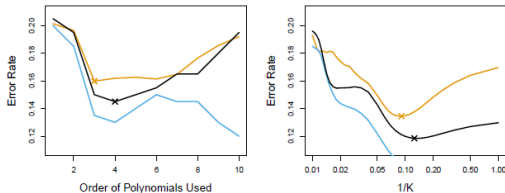


FIGURE 5.8. Test error (brown), training error (blue), and 10-fold CV error (black) on the two-dimensional classification data displayed in Figure 5.7. Left: Logistic regression using polynomial functions of the predictors. The order of the polynomials used is displayed on the x-axis. Right: The KNN classifier with different values of K , the number of neighbors used in the KNN classifier.

Reading the figure

- ▶ Blue: training error.
- ▶ Brown: true test error.
- ▶ Black: 10-fold CV error.
- ▶ Training error keeps falling with flexibility; test error is U-shaped.

Do not select by training error

Training error always prefers the most flexible model. CV error tracks generalisation instead.

The Bootstrap: Main Idea

Definition

The **bootstrap** estimates the uncertainty of a statistic by repeatedly sampling *with replacement* from the empirical distribution $\hat{F}$ of the observed data.

Observed sample

$\mathcal{Z} = \{z_1, \dots, z_N\}$. Each bootstrap sample $\mathcal{Z}^*$ also has size N , drawn with replacement from $\mathcal{Z}$.

What it estimates

- ▶ Standard error of a statistic.
- ▶ Variability of model coefficients.
- ▶ Uncertainty when formulas are unavailable.

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Motivating Example: Investment Allocation

Problem

Invest fraction α in asset X , $1 - \alpha$ in asset Y . Choose α to minimise the portfolio variance $\text{Var}(\alpha X + (1 - \alpha)Y)$.

Optimal α

Differentiating the quadratic $\alpha^2\sigma_X^2 + (1 - \alpha)^2\sigma_Y^2 + 2\alpha(1 - \alpha)\sigma_{XY}$ and solving gives

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}. \quad (5.5)$$

The catch

$\sigma_X^2, \sigma_Y^2, \sigma_{XY}$ are unknown — we plug in sample estimates $\hat{\alpha}$, but $\hat{\alpha}$ itself varies from sample to sample.

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Figure: $\hat{\alpha}$ Changes Across Samples

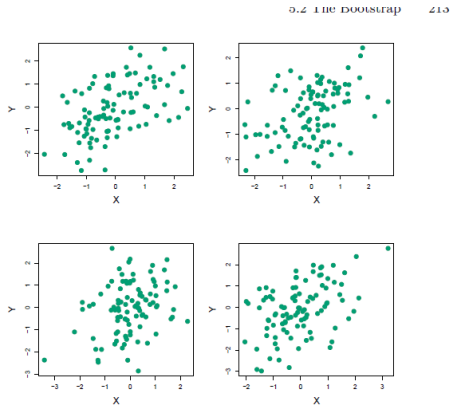


FIGURE 5.9. Each panel displays 100 simulated returns for investments X and Y . From left to right and top to bottom, the resulting estimates for α are 0.576, 0.532, 0.657, and 0.651.

Reading the figure

- ▶ Each panel: 100 simulated (X, Y) pairs.
- ▶ Each sample gives a different $\hat{\alpha}$.

Problem

In real data we cannot repeatedly resample from the true population to see this variability directly.

Bootstrap Sampling with Replacement

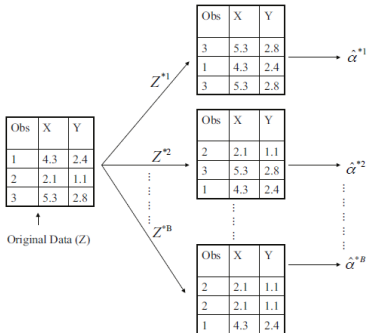


FIGURE 5.11. A graphical illustration of the bootstrap approach on a small sample containing $n = 3$ observations. Each bootstrap data set contains n observations, sampled with replacement from the original data set. Each bootstrap data set is used to obtain an estimate of α .

The trick

- ▶ Start with the original data $\mathcal{Z}$.
- ▶ Draw N observations *with replacement*.
- ▶ Some points appear multiple times, some not at all.
- ▶ Compute the statistic on each bootstrap sample.

Bootstrap Algorithm

Given $\mathcal{Z} = \{z_1, \dots, z_N\}$

1. Draw $\mathcal{Z}^{*1}$ by sampling N points from $\mathcal{Z}$ with replacement.
2. Compute $\hat{\alpha}^{*1} = s(\mathcal{Z}^{*1})$.
3. Repeat B times, obtaining $\hat{\alpha}^{*1}, \dots, \hat{\alpha}^{*B}$.
4. Estimate the standard error from the spread of these estimates.

Bootstrap principle

$\text{SE}\{S(\mathcal{Z})\} = \sqrt{\text{Var}_F\{S(\mathcal{Z})\}}$ is approximated by replacing the unknown F with $\hat{F}$:

$$\text{SE}\{S(\mathcal{Z})\} \approx \sqrt{\text{Var}_{\hat{F}}\{S(\mathcal{Z}^*)\}}.$$

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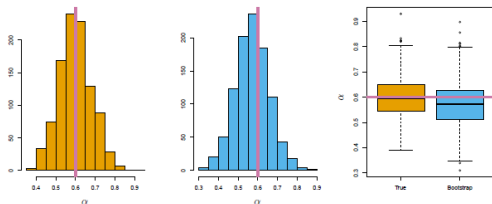
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Bootstrap Standard Error Formula

Bootstrap SE

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B \left(\hat{\alpha}^{*r} - \frac{1}{B} \sum_{r'=1}^B \hat{\alpha}^{*r'} \right)^2}. \quad (5.6)$$

- ▶ B : number of bootstrap samples; $\hat{\alpha}^{*r}$: estimate from bootstrap sample r .
- ▶ This is just the sample standard deviation of the bootstrap estimates.



Left: true sampling distribution (many population draws). Middle: bootstrap distribution (one sample, resampled). Right: boxplots showing similar spread — the bootstrap mimics the sampling distribution using only observed data.

FIGURE 5.10. Left: A histogram of the estimates of α obtained by generating 1,000 simulated data sets from the true population. Center: A histogram of the estimates of α obtained from 1,000 bootstrap samples from a single data set. Right: The estimates of α displayed in the left and center panels are shown as

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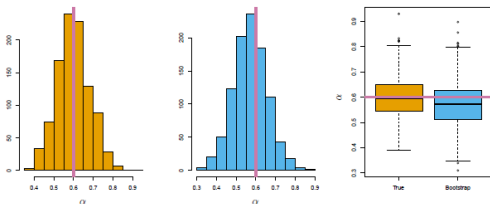


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Cross-Validation vs. Bootstrap

Method	Main purpose	Resampling style
Cross-validation	estimate prediction error	train/validation splits
Bootstrap	estimate estimator variability	sampling <i>with</i> replacement from $\hat{F}$

Do not confuse the goals

Use cross-validation to assess or select *predictive models*. Use the bootstrap to quantify the *uncertainty of an estimator* — almost any statistic, as long as you can recompute it on resampled data.

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In Practice: K -Fold CV in a Few Lines

```
import numpy as np
from sklearn.model_selection import (
    KFold, cross_val_score)
from sklearn.linear_model import LinearRegression

kf = KFold(n_splits=10, shuffle=True,
           random_state=0)
model = LinearRegression()

scores = cross_val_score(
    model, X, y, cv=kf,
    scoring="neg_mean_squared_error")

cv_mse = -scores.mean()
print(f"10-fold CV MSE: {cv_mse:.3f}")
```

What is happening

KFold builds the K train/validation splits; `cross_val_score` refits the model K times and returns one score per fold — exactly Equation (5.3).

Habit

Never report a single train/test split's number as “the” test error — report the CV mean (and, if it matters, its spread across folds).

Summary & What Comes Next

Key takeaways

- ▶ Cross-validation estimates test error *without* permanently sacrificing a held-out set: $K = 5$ or 10 is the usual bias–variance compromise, LOOCV is the low-bias, higher-variance extreme.
- ▶ The bootstrap estimates the variability of *any* statistic by resampling with replacement from the observed data.
- ▶ Together, they let us evaluate and compare models honestly using only the sample we already have.

Next session: Model Selection and Regularization

Now that we can estimate test error reliably, we can *use* that estimate: to choose between competing models, and to control model complexity directly via shrinkage — **ridge regression** and the **lasso**.

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Resampling Methods

Reusing data carefully to estimate generalisation and uncertainty

Questions & discussion welcome

